The skeleton in the skew-symmetrizable acyclic case

Cluster skeleton geometry: formalization reference

This document develops the supplied skeleton argument for an integral unimodular completion E and arbitrary positive rational diagonal $\widetilde{D}$ with $E\widetilde{D}$ skew-symmetric. The weights are part of the theorem, not shorthand for $\widetilde{D} = I$. The existing Lean library verifies only that specialization; formalization of the weighted theorem below is the current task.

1 Skeleton in the acyclic case

Let M and F be the mutable and frozen index sets, and put $m = |M|$, $f = |F|$ and $n = m + f$. Fix an integral completion $E \in GL_n(\mathbb{Z})$ whose mutable columns are $\widetilde{B}$. Fix a positive rational diagonal matrix $\widetilde{D} = \mathrm{diag}(d_j)$ such that $K = E\widetilde{D}$ is skew-symmetric. The skew-symmetric case is $\widetilde{D} = I_n$ and $K = E$. The mutable block $B = E_{M,M}$ is acyclic: there is no directed cycle with all successive matrix entries positive. Frozen coordinates remain invertible variables. The unimodular completion hypothesis is retained throughout.

Write $c_i = Ee_i$ and $z^{c_i} = \prod_j z_j^{E_{ji}}$. On the initial torus put

$$\Omega = \frac{1}{2}\sum_{j,l} K_{jl}\, d\log z_j \wedge d\log z_l, \qquad \lambda_{\mathrm{tor}} = u^{\mathsf{T}} K\, d\theta.$$

The extension and nondegeneracy of Ω follow from the node-chart calculation below; they are construction obligations, not extra input. For a complete Liouville field, define its dynamical skeleton as the set of points with relatively compact forward orbit. Our sign convention is $\iota_Z d\lambda = \lambda$.

A subset $I \subset M$ is *independent* if $B_{ij} = 0$ for all $i, j \in I$.

In the acyclic case, $\mathcal{A}(\widetilde{B})$ can be presented by generators and relations. For $i \in M$ put

$$P_i^{\pm} = \prod_{j \in M \sqcup F} z_j^{[\pm \widetilde{B}_{ji}]_+}, \qquad t_i = P_i^+ / P_i^- = z^{c_i}.$$

By [1, Corollary 1.21],

$$\mathcal{A}(\widetilde{B}) = \frac{\mathbb{C}[z_i, z_i' \ (i \in M),\ z_j^{\pm 1} \ (j \in F)]}{(z_i z_i' - P_i^+ - P_i^-)_{i \in M}}. \tag{1}$$

Lemma 1. *We have the following.*

(i) *At every point $p \in \mathcal{A}(\widetilde{B})$, the set $\{i \in M \mid z_i(p) = 0\}$ is independent.*

(ii) *Let I be independent, and let $U_I = \{z_j \neq 0 \ (j \in M \backslash I)\} \subset \mathcal{A}(\widetilde{B})$. Then $U_I \cong \mathcal{N}^I \times (\mathbb{C}^*)^{n-2|I|}$, where the ith factor $\mathcal{N}$ has coordinates*

$$(x, y, t) = (z_i,\ z_i'/P_i^-,\ t_i) \qquad (i \in I).$$

(iii) *The open sets U_I cover $\mathcal{A}(\widetilde{B})$.*

Proof. (i) Fix a point p, let $Z = \{i \in M : z_i(p) = 0\}$, and suppose that $B_{Z,Z} \neq 0$. Since B is acyclic, we can take a maximal path $j_1, \dots, j_r$ in Z with $r \geq 2$ and $B_{j_s j_{s+1}} > 0$. Put $i = j_r$. By maximality, $B_{ji} \geq 0$ for all $j \in Z$, and $B_{j_{r-1} i} > 0$. So $P_i^+(p) = 0$ since $z_{j_{r-1}}(p) = 0$. Every mutable j in P_i^- has $B_{ji} < 0$, so $j \notin Z$; the frozen z_j are units. So $P_i^-(p) \neq 0$. This contradicts the relation $z_i z_i' = P_i^+ + P_i^-$.

(ii) On U_I, the z_j with $j \in (M \sqcup F) \setminus I$ are units. For $i \in I$, the monomials $P_i^\pm$ involve only these z_j. For $j \in M \setminus I$, we have $z_j' = (P_j^+ + P_j^-)/z_j$. So (1) gives

$$\mathcal{O}(U_I) = \mathbb{C}[z_j^{\pm 1} \ (j \in M \sqcup F \setminus I)]\,[z_i, y_i \ (i \in I)]/(z_i y_i - 1 - t_i)_{i \in I}, \qquad y_i = z_i'/P_i^-.$$

The vectors c_i, $i \in I$, are columns of $E \in GL_n(\mathbb{Z})$ and lie in $\mathbb{Z}^{(M \sqcup F) \setminus I}$. So they extend to a basis of $\mathbb{Z}^{(M \sqcup F) \setminus I}$, which implies $\mathbb{C}[z_j^{\pm 1} \ (j \notin I)] = \mathbb{C}[t_i^{\pm 1} \ (i \in I)] \otimes \mathcal{O}((\mathbb{C}^*)^{n-2|I|})$. This proves (ii).

(iii) For any point $p \in \mathcal{A}(\widetilde{B})$, take $I = \{i \in M \mid z_i(p) = 0\}$. Then $p \in U_I$. □

Weighted form on the actual node charts

Positivity of the d_i implies $E_{ii} = 0$ and $E_{ij} = 0 \iff E_{ji} = 0$, with opposite signs when nonzero. These are the sign properties used in the zero-set proof. The columns c_i for independent I span a direct summand of the complement coordinate lattice: the integer inverse of E restricts to a retraction. Choose complementary characters $s_1, \dots, s_{n-2|I|}$. In logarithmic coordinates (x_i, t_i, s_a) the form has the shape

$$\Omega = \sum_{i \in I} d_i \, d\log t_i \wedge d\log x_i + \Omega_0(t, s),$$

where Ω_0 has constant coefficients in $d\log t_i, d\log s_a$. Indeed the x_i row is determined by $\sum_j K_{ij} d\log z_j = -d_i d\log t_i$. Its skew matrix, in these coordinates, is

$$\begin{pmatrix} 0 & -D_I & 0 \\ D_I & A & B_0 \\ 0 & -B_0^{\mathsf{T}} & C \end{pmatrix}.$$

It is nonsingular, so C is nonsingular, by eliminating the first two block rows (with the empty-block convention when necessary). On the node $x_i y_i = 1 + t_i$,

$$d\log t_i \wedge d\log x_i = t_i^{-1} \, dy_i \wedge dx_i.$$

This extends across $x_i = 0$. At any given point choose I to be its zero set. There $t_i = -1$ and $dt_i = y_i dx_i$. Thus the residual dt_i terms contribute only $dx_i \wedge dx_j$ and $dx_i \wedge d\log s_a$ terms. Pairing against the y_i directions, then the s_a directions, then the x_i directions proves nondegeneracy, using $d_i \neq 0$ and nonsingularity of C. The local extensions agree on overlaps by density of the initial torus. The node charts are smooth, and their coordinate maps are the actual exchange coordinates; they give the smooth atlas, dense torus, and immersive coordinate inclusion used in the target theorem.

Cutoffs and the local node

The node is $\mathcal{N} = \{(x, y, t) \in \mathbb{C}^2 \times \mathbb{C}^* : xy = 1 + t\}$. On $x \neq 0$ write $x = e^{u+i\theta}$ and $t = e^{v+i\eta}$. Its holomorphic symplectic form is $d\log t \wedge d\log x = dy \wedge dx/(xy - 1)$. Choose $L > a > 0$ and

smooth functions satisfying

$$h \le 0, \quad 0 \le h' \le 1, \quad u - h(u) < 0 \ (u < 0),$$
$$h(u) = u + a \ (u \le -L), \qquad h = 0 \text{ near } [0, \infty),$$
$$\rho \in C_c^\infty(\mathbb{R}, [0,1]), \qquad \rho = 1 \text{ near } 0,$$
$$\beta \in C^\infty(\mathbb{R}/2\pi\mathbb{Z}), \quad \operatorname{supp}\beta \subset (\pi - \varepsilon, \pi + \varepsilon),$$
$$\beta(\eta) = \eta - \pi \text{ near } \pi, \quad (\eta - \pi)\beta(\eta) \ge 0, \quad \beta' \le 1, \qquad 0 < \varepsilon < \pi.$$

The angular difference is the lift in $(-\varepsilon, \varepsilon)$ on this arc. The node primitive is $v\,d\theta - u\,d\eta + d(h\rho\beta)$. These cutoff properties, including existence, are part of the formalization.

Theorem 1. *There is a Liouville form λ_{ac} of $\Im\Omega$ on $\mathcal{A}(\widetilde{B})$ such that*

$$\Lambda = \operatorname{Skel}(Z_{\lambda_{\mathrm{ac}}}) = \bigcup_{I \subset M \text{ independent}} \Lambda_I,$$
$$\Lambda_I = \{z_i' = 0, \ |z_i| \le 1 \ (i \in I), \quad |z_j| = 1 \ (j \in (M \sqcup F) \setminus I)\}. \tag{2}$$

Proof. Use the constants $L > a > 0$ and the cutoffs h, ρ, β specified above. Let $\lambda_{\mathrm{tor}} = u^{\mathsf{T}} K \mathrm{d}\theta$. Fix $\delta_0 > 0$ with $h = 0$ on $(-\delta_0, \infty)$. Choose a smooth nonincreasing $\vartheta\colon \mathbb{R} \to [0,1]$ with $\vartheta = 1$ on $(-\infty, -1]$ and $\vartheta = 0$ on $[0, \infty)$. On the initial torus $(\mathbb{C}^*)^n$ write

$$u_j = \log|z_j|, \qquad \theta_j = \arg z_j, \qquad v_i = \log|t_i| = c_i(u), \qquad \eta_i = \arg t_i = c_i(\theta),$$

and set

$$\chi_i := \prod_{j \in M:\, B_{ji} \ne 0} \vartheta(u_i - u_j), \qquad H := \sum_{i \in M} d_i\, h(u_i)\, \chi_i\, \rho(v_i)\, \beta(\eta_i), \qquad \lambda_{\mathrm{ac}} := \lambda_{\mathrm{tor}} + \mathrm{d}H.$$

An empty product is 1. Note that if $B_{ij} \ne 0$, then $\chi_i \chi_j = 0$.
Let $I \subset M$ be independent. For $i, i' \in I$, $K_{ii'} = d_{i'} B_{ii'} = 0$. For $i \in M$,

$$\sum_j K_{ji}\, u_j = d_i \sum_j E_{ji} u_j = d_i v_i, \qquad \sum_l K_{il}\, \mathrm{d}\theta_l = -\sum_l K_{li}\, \mathrm{d}\theta_l = -d_i\, \mathrm{d}\eta_i.$$

The terms of λ_{tor} with $l \in I$ give $\sum_{i \in I} d_i v_i\, \mathrm{d}\theta_i$. The terms with $j \in I$ and $l \notin I$ give $-\sum_{i \in I} d_i u_i\, \mathrm{d}\eta_i$, since $K_{il} = 0$ for $i, l \in I$. Hence

$$\lambda_{\mathrm{tor}} = \sum_{i \in I} d_i (v_i\, \mathrm{d}\theta_i - u_i\, \mathrm{d}\eta_i) + \sum_{j,l \in (M \sqcup F) \setminus I} K_{jl}\, u_j\, \mathrm{d}\theta_l. \tag{3}$$

We first show that λ_{ac} is globally defined. Let $q \in \mathcal{A}(\widetilde{B}) \setminus (\mathbb{C}^*)^n$, and let $I = \{i \in M \mid z_i(q) = 0\}$. Then $I \ne \emptyset$, I is independent by Lemma 1(i), and $q \in U_I$. Let $i \in I$. The monomials $P_i^{\pm}$ only involve z_j with $j \notin I$, so they are units on U_I. Hence $t_i = P_i^+ / P_i^-$ is a unit on U_I, and $0 = z_i(q) z_i'(q) = P_i^-(q)\big(1 + t_i(q)\big)$ gives $t_i(q) = -1$. So, as $p \to q$ with $p \in (\mathbb{C}^*)^n$,

$$u_i(p) \to -\infty, \qquad v_i(p) \to 0, \qquad \eta_i(p) \to \pi, \qquad u_j(p) \to \log|z_j(q)| \in \mathbb{R} \quad \big(j \in (M \sqcup F) \setminus I\big).$$

Choose a neighborhood $W \subset U_I$ of q such that, for $i \in I$ and $j \in M$ with $B_{ji} \ne 0$,

$$\rho(v_i) = 1, \quad \beta(\eta_i) = \eta_i - \pi \quad \text{on } W, \qquad u_i \le -L, \quad u_i \le u_j - 1 \quad \text{on } W \cap (\mathbb{C}^*)^n.$$

Then, on $W \cap (\mathbb{C}^*)^n$, $h(u_i) = u_i + a$ and $\chi_i = 1$. Also $\chi_j = 0$ there, because χ_j has the factor $\vartheta(u_j - u_i)$ and $u_j - u_i \ge 1$. So, on $W \cap (\mathbb{C}^*)^n$,

$$H = \sum_{i \in I} d_i (u_i + a)(\eta_i - \pi) + H', \tag{4}$$

where H' is the sum of the summands of H with index $k \in M \setminus I$ such that $B_{ik} = 0$ for all $i \in I$. For such k, the functions u_k, χ_k, $v_k = c_k(u)$ and $\eta_k = c_k(\theta)$ only involve z_j with $j \notin I$, by the support of c_k. These z_j are units on U_I. So H' does not depend on u_i and θ_i, $i \in I$.

By (3) and (4), on $W \cap (\mathbb{C}^*)^n$,

$$\lambda_{\mathrm{ac}} = \sum_{j,l\notin I} K_{jl}\, u_j \,\mathrm{d}\theta_l + \mathrm{d}H' + \sum_{i\in I} d_i(v_i \,\mathrm{d}\theta_i + (\eta_i - \pi)\,\mathrm{d}u_i + a\,\mathrm{d}\eta_i).$$

Here we used $v_i\,\mathrm{d}\theta_i - u_i\,\mathrm{d}\eta_i + \mathrm{d}((u_i+a)(\eta_i-\pi)) = v_i\,\mathrm{d}\theta_i + (\eta_i-\pi)\,\mathrm{d}u_i + a\,\mathrm{d}\eta_i$. The first two terms are smooth on W. On W, $1 - z_i y_i = -t_i$ and $\mathrm{Log}(-t_i) = v_i + \mathrm{i}(\eta_i - \pi)$. So, by the holomorphic logarithm near 1,

$$v_i\,\mathrm{d}\theta_i + (\eta_i-\pi)\,\mathrm{d}u_i + a\,\mathrm{d}\eta_i = \Im\Big(\frac{\mathrm{Log}(1-z_iy_i)}{z_i}\,\mathrm{d}z_i\Big) + a\,\mathrm{d}\arg(1-z_iy_i).$$

The right side is smooth on W, because $\mathrm{Log}(1-z_iy_i)/z_i$ is holomorphic on W, with value $-y_i$ on $\{z_i = 0\}$. Hence λ_{ac} extends smoothly over W, hence also extends to $\mathcal{A}(\widetilde{B})$.

Now we consider the Liouville field $Z = Z_{\lambda_{\mathrm{ac}}}$, given by $\iota_Z \Im\Omega = \lambda_{\mathrm{ac}}$. A dot denotes the derivative along Z. On the initial torus, $\Im\Omega = \sum_{j,l} K_{jl}\,\mathrm{d}u_j \wedge \mathrm{d}\theta_l$. Comparing the coefficients of $\mathrm{d}\theta$ and of $\mathrm{d}u$ in $\iota_Z\Im\Omega = \lambda_{\mathrm{tor}} + \mathrm{d}H$ gives

$$K^{\mathsf{T}}\dot{u} = K^{\mathsf{T}}u + \partial_\theta H, \qquad -K\dot{\theta} = \partial_u H.$$

The function H depends on θ only through $\eta_i = c_i(\theta)$, and $d_i c_i = K e_i$. So

$$\partial_\theta H = \sum_{i\in M} d_i\, h(u_i)\chi_i\rho(v_i)\beta'(\eta_i)\, c_i = K \sum_{i\in M} h(u_i)\chi_i\rho(v_i)\beta'(\eta_i)\, e_i.$$

Since $K^{\mathsf{T}} = -K$, this gives

$$\begin{aligned} \dot{u}_j &= u_j && (j \in F),\\ \dot{u}_i &= u_i - h(u_i)\,\chi_i\,\rho(v_i)\,\beta'(\eta_i) && (i \in M),\\ \dot{\theta} &= -K^{-1}\partial_u H. \end{aligned} \tag{5}$$

For $i \in M$, $c_i(K^{-1}x) = -x_i/d_i$, because $d_i c_i^{\mathsf{T}} = e_i^{\mathsf{T}} K^{\mathsf{T}} = -e_i^{\mathsf{T}} K$. Also $c_i(e_k) = E_{ki} = B_{ki}$ for $k \in M$. So applying c_i to (5) gives

$$\dot{\eta}_i = \frac{1}{d_i}\,\partial_{u_i} H, \qquad \dot{v}_i = v_i - \sum_{k\in M} B_{ki}\, h(u_k)\,\chi_k\,\rho(v_k)\,\beta'(\eta_k) \qquad (i \in M). \tag{6}$$

We now bound $\dot{u}$. Since $h \le 0$, $0 \le \chi_i\rho \le 1$ and $\beta' \le 1$, we have $h\chi_i\rho\beta' \ge h\chi_i\rho \ge h$. With $h = 0$ on $(-\delta_0, \infty)$, (5) gives, for $i \in M$,

$$\dot{u}_i \le u_i - h(u_i) \quad \text{on } \{u_i < 0\}, \qquad \dot{u}_i = u_i \quad \text{on } \{u_i > -\delta_0\}. \tag{7}$$

Recall that $u - h(u) < 0$ for $u < 0$, and $u - h(u) = -a$ for $u \le -L$. Also $|h(u)| \le |u| + L$, since $h(u) = u + a$ for $u \le -L$, and $a - L \le h \le 0$ on $[-L, \infty)$. So (5) gives $|\dot{u}_j| \le C(1 + |u_j|)$ for all $j \in M \sqcup F$, with $C = (1 + \max|\beta'|)(1 + L)$. By density, (7) holds on $\{z_i \ne 0\}$, and the bound holds on $\{z_j \ne 0\}$.

Next we compute Z near $\{z_i = 0\}$. Let q, I and W be as above, and let $i \in I$. On $W \cap (\mathbb{C}^*)^n$, $h(u_i) = u_i + a$, $\chi_i = \rho(v_i) = \beta'(\eta_i) = 1$, and $\chi_k = 0$ for $B_{ki} \ne 0$. Also $\partial_{u_i} H = d_i(\eta_i - \pi)$ there, by (4). So (5) and (6) give

$$\dot{u}_i = -a, \qquad \dot{v}_i = v_i, \qquad \dot{\eta}_i = \eta_i - \pi \qquad \text{on } W \cap (\mathbb{C}^*)^n.$$

Since $\operatorname{Log}(-t_i) = v_i + \mathrm{i}(\eta_i - \pi)$, by density

$$Z|z_i|^2 = -2a|z_i|^2, \qquad \dot{t}_i = t_i \operatorname{Log}(-t_i) \qquad \text{on } W. \tag{8}$$

Every point of $\{z_i = 0\}$ is such a q, so these sets W cover $\{z_i = 0\}$. Along an orbit near $\{z_i = 0\}$, $|z_i|^2$ solves $\frac{\mathrm{d}}{\mathrm{d}\tau}|z_i|^2 = -2a|z_i|^2$. By uniqueness of solutions, $\{z_i = 0\}$ and its complement are invariant under the flow of Z.

On $\{z_i = 0\}$, the z_j with $B_{ji} \neq 0$ are nonzero, by Lemma 1(i), so $P_i^- \neq 0$. So $y_i = z_i'/P_i^-$ is defined near $\{z_i = 0\}$, and $y_i = (1 + t_i)/z_i$ where $z_i \neq 0$. Put $\phi(t) = t \operatorname{Log}(-t)/(1 + t)$. It is holomorphic near $t = -1$, with $\phi(-1) = 1$. On $W \cap \{z_i y_i \neq 0\}$, (8) gives

$$Z \log|y_i| = \Re \frac{\dot{t}_i}{1 + t_i} - Z \log|z_i| = \Re\phi(t_i) + a.$$

By density, $Z|y_i|^2 = 2(\Re\phi(t_i) + a)|y_i|^2$ on W. On $W \cap \{z_i = 0\}$, $t_i = -1$, so

$$Z|y_i|^2 = 2(1 + a)|y_i|^2 \qquad \text{on } \{z_i = 0\}. \tag{9}$$

Hence $|y_i(\phi_Z^\tau(p))| = e^{(1+a)\tau}|y_i(p)|$ for $p \in \{z_i = 0\}$ and all τ.

Next we show that Z is complete. Let $\gamma\colon (\tau_-, \tau_+) \to \mathcal{A}(\widetilde{B})$ be a maximal integral curve of Z, and let $I_0 = \{i \in M \mid z_i(\gamma(0)) = 0\}$. By invariance, $z_i = 0$ on γ for $i \in I_0$, and $z_j \neq 0$ on γ for $j \in (M \sqcup F) \setminus I_0$. Let $T > 0$. For $\tau \in (\tau_-, \tau_+)$ with $|\tau| \leq T$, the bound $|\dot{u}_j| \leq C(1 + |u_j|)$ and (9) give

$$|u_j(\gamma(\tau))| \leq (1 + |u_j(\gamma(0))|)e^{CT} \quad (j \notin I_0), \qquad |y_i(\gamma(\tau))| \leq e^{(1+a)T}|y_i(\gamma(0))| \quad (i \in I_0).$$

So the coordinates $z_j^{\pm 1}$ $(j \notin I_0)$, $z_j' = (P_j^+ + P_j^-)/z_j$ $(j \in M \setminus I_0)$ and $z_i' = P_i^- y_i$ $(i \in I_0)$ are bounded on these $\gamma(\tau)$. By (1), $\mathcal{A}(\widetilde{B})$ is closed in $\mathbb{C}^{2m} \times (\mathbb{C}^*)^f$, so these $\gamma(\tau)$ lie in a compact set. Hence $\tau_\pm = \pm\infty$, since a maximal integral curve with $\tau_+ < \infty$ or $\tau_- > -\infty$ leaves every compact set.

We now prove (2).

(i) $\operatorname{Skel}(Z) \subset \cup_I \Lambda_I$. Let $p \in \operatorname{Skel}(Z)$, so that the closure of $\gamma(\tau) = \phi_Z^\tau(p)$, $\tau \geq 0$, is compact. Put

$$I_0 = \{i \in M \mid z_i(p) = 0\}, \qquad I = I_0 \cup \{i \in M \mid z_i(p) \neq 0,\ u_i(p) < 0\}.$$

We show that $p \in \Lambda_I$.

We first determine u along γ. For $j \in F$, $u_j(\gamma(\tau)) = e^\tau u_j(p)$ is bounded, so $u_j(p) = 0$. Let $i \in M \setminus I_0$. If $u_i(p) > 0$, then $u_i(\gamma(\tau)) = e^\tau u_i(p)$ by (7), which is unbounded. If $u_i(p) = 0$, then $u_i = 0$ on γ, by (7) and uniqueness for $\dot{u}_i = u_i$. If $u_i(p) < 0$, then u_i decreases along γ, and (7) gives

$$\dot{u}_i \leq \sup_{u \leq u_i(p)} \big(u - h(u)\big) < 0 \qquad \text{on } \gamma.$$

The supremum is negative because $u - h(u) < 0$ for $u < 0$ and $u - h(u) = -a$ for $u \leq -L$. So $u_i(\gamma(\tau)) \to -\infty$. In summary,

$$u_j = 0 \text{ on } \gamma \quad (j \in (M \sqcup F) \setminus I), \qquad u_i(\gamma(\tau)) \to -\infty \quad (i \in I \setminus I_0), \qquad z_i = 0 \text{ on } \gamma \quad (i \in I_0).$$

Let q be a limit point of γ. Then $\{i \in M \mid z_i(q) = 0\} = I$, so I is independent by Lemma 1(i). Also $|z_j| = 1$ on γ and at q for $j \in M \setminus I$. So the closure of γ is a compact subset of U_I. On U_I, $z_i' = P_i^- y_i$ with P_i^- a unit, for $i \in I$. Since $|z_i(p)| \leq 1$ for $i \in I$ and $|z_j(p)| = 1$ for $j \notin I$, it remains to show that $y_i(p) = 0$ for $i \in I$.

Let $i \in I_0$. By (9), $|y_i(\gamma(\tau))| = e^{(1+a)\tau}|y_i(p)|$, which is bounded only if $y_i(p) = 0$.

Let $i \in I \setminus I_0$. We show that $t_i = -1$ on γ. The character c_i is supported on $F \sqcup \{j \in M \mid B_{ji} \neq 0\}$, which is disjoint from I. So $v_i = c_i(u) = 0$ on γ. Consider the open set

$$O = \{z_i \neq 0,\ |z_j| > e^{-\delta_0}\ (j \in M,\ B_{ji} \neq 0)\} \supset \gamma.$$

Besides the ith summand of H, only the summands with index j, $B_{ji} \neq 0$, depend on u_i, through χ_j and $v_j = c_j(u)$. These vanish on $O \cap (\mathbb{C}^*)^n$, because $h(u_j) = 0$ there. Also $\partial_{u_i} v_i = E_{ii} = 0$. So (6) gives, on $O \cap (\mathbb{C}^*)^n$, and then on O by density,

$$\dot{\eta}_i = \big(h'(u_i)\,\chi_i + h(u_i)\,\partial_{u_i}\chi_i\big)\,\rho(v_i)\,\beta(\eta_i).$$

The factor in front of $\beta(\eta_i)$ is ≥ 0, because $h' \geq 0$, $h \leq 0$, $\chi_i \geq 0$, $\rho \geq 0$, and $\partial_{u_i}\chi_i \leq 0$ as $\vartheta' \leq 0$. Since $(\eta - \pi)\beta(\eta) \geq 0$ and $\operatorname{supp}\beta \subset (\pi - \varepsilon, \pi + \varepsilon)$, the distance $|\eta_i - \pi|$ in $\mathbb{R}/2\pi\mathbb{Z}$ is nondecreasing along γ. On the other hand, $z_i(\gamma(\tau)) \to 0$, and y_i is bounded on γ. So $1 + t_i = z_i y_i \to 0$, and $|\eta_i - \pi| \to 0$. Hence $\eta_i = \pi$ on γ. With $v_i = 0$, this gives $t_i = -1$ on γ, so $y_i = (1 + t_i)/z_i = 0$ on γ. Thus $p \in \Lambda_I$, and $\mathrm{Skel}(Z) \subset \bigcup_I \Lambda_I$.

(ii) $\Lambda_I \subset \mathrm{Skel}(Z)$.

Let I be independent, and let $p \in \Lambda_I$. The set Λ_I is closed, and on it the coordinates $z_j^{\pm 1}$ $(j \notin I)$, z_i, $z_i' = 0$ $(i \in I)$ and $z_j' = (P_j^+ + P_j^-)/z_j$ $(j \in M \setminus I)$ are bounded. So Λ_I is compact, and it suffices to show $\phi_Z^\tau(p) \in \Lambda_I$ for $\tau \geq 0$. Let $\gamma(\tau) = \phi_Z^\tau(p)$. As above, $u_j = 0$ on γ for $j \notin I$. For $i \in I$, $z_i = 0$ on γ if $z_i(p) = 0$, and $u_i \leq 0$ on γ if $z_i(p) \neq 0$. It remains to show $y_i = 0$ on γ for $i \in I$. If $z_i(p) = 0$, this follows from (9) and $y_i(p) = 0$. If $z_i(p) \neq 0$, then $v_i = 0$ on γ, and η_i solves the equation for $\dot{\eta}_i$ above, with $\eta_i(p) = \pi$. Since $\beta(\pi) = 0$, uniqueness gives $\eta_i = \pi$ on γ. So $t_i = -1$ and $y_i = 0$ on γ. This proves (2). □

Formalization target and convention matching

All geometry is on the actual complex points of the exchange equations with their induced topology. Construct the smooth presentation, the extended symplectic form, the primitive, and a jointly smooth flow defined for every real time. Define the coordinate cap and dynamical trapped set independently, and prove their equality for this same flow. The equivalent coordinate cap is

$$\{p : |z_f| = 1\ (f \in F),\ |z_i| \leq 1,\ (1 - |z_i|^2)z_i' = 0\ (i \in M)\}.$$

For the converse equivalence, the indices with $|z_i| < 1$ have $z_i' = 0$; the zero-set argument applied to their exchange relations makes them independent (equivalently use the existing strict-cap-independence lemma).

Current formalization obligation. The completed baseline uses a skew-symmetric integral seed and verifies the case $\widetilde{D} = I$ only. Its cap, strata, chart-rank and same-flow equality lemmas are reusable starting points. Their old verification receipts do not verify the weighted theorem above. The new goal quantifies over every integral unimodular E and every positive rational diagonal $\widetilde{D}$ satisfying $E\widetilde{D} + (E\widetilde{D})^{\mathsf{T}} = 0$. Carry the weights through the atlas, two-form, primitive, vector field, completeness and both skeleton inclusions. In particular, $d_i c_i = K e_i$ and $c_i^{\mathsf{T}} K^{-1} = -e_i^{\mathsf{T}}/d_i$ explain the cancellations in the flow equations.

The old row-exponent convention agrees with the present column convention by monomial interchange only when $\widetilde{D} = I$. For general weights, use column exponents or supply a proved convention equivalence; never replace the exchange exponents by entries of K. Require the explicit non-skew input in the target PDF, as well as the theorem for all inputs and recovery of the $\widetilde{D} = I$ specialization.

The real smooth Liouville/skeleton theorem is the formalization target. Use native Mathlib geometry and the pinned FsFltz interfaces. Construct all witnesses rather than assuming a

symplectic extension or complete flow. Final verification must compile the entire generalized library, check theorem consumers and transitive axioms, and bind new receipts to the exact source. Only `propext`, `Classical.choice` and `Quot.sound` are permitted. No principal-coefficient restriction, extra geometric axiom, or specialization $d_i = 1$ may be introduced. Unimodularity and mutable acyclicity remain hypotheses; categorical HMS is outside this task.

References

[1] A. Berenstein, S. Fomin, A. Zelevinsky, *Cluster algebras III: Upper bounds and double Bruhat cells*, Duke Math. J. 126 (2005), 1–52, Corollary 1.21.